

Mind the Gap:



How Proximity to Metro Stops Relates to Gentrification of Local Neighborhoods

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Key Findings:

Gentrification Risk

01

Gentrification is negatively related to Metro proximity

Gentrification risk increases steadily as distance to the nearest station decreases across the sample.

02

Risk levels differ by Metro Line

Risk levels differ sharply from one Metro line to the next as well (Kruskal-Wallis $p < 0.001$), which is why the forecast trains on the corridors most like the D Line rather than the full network.

03

Pressure shifts to Koreatown, Palms, and Mar Vista

Our analysis predicts little to no added pressure in the Westwood and UCLA corridor, since rents there have already topped out, with little room left to climb. Pressure shifts to still-affordable areas like Koreatown, Palms, and Mar Vista, which carry the highest composite gentrification risk along the line.



Background



01

Los Angeles is extending its metro rail with the D Line, and transit proximity is routinely marketed as a neighborhood amenity.

02

That same amenity has coincided with rising rents and displacement along earlier rail expansions across LA, and the same trend has been documented in other cities as well.

03

With 63% of Angelenos renting, a shift in housing costs reaches most of the population, not a narrow slice of it. The question this analysis takes up is where, along the new corridor, that pressure is most likely to land.



Approach



Outcome

- ❑ We computed a Composite Gentrification Risk Score (CGRS) that captures:
 - ❑ **rent premium,**
 - ❑ **rent burden,**
 - ❑ **spatial rent appreciation,**
 - ❑ **price per square foot premium,**
 - ❑ **& education change.**
- ❑ Each component is z-scored and averaged with equal weight, so the final CGRS places every listing on a common scale

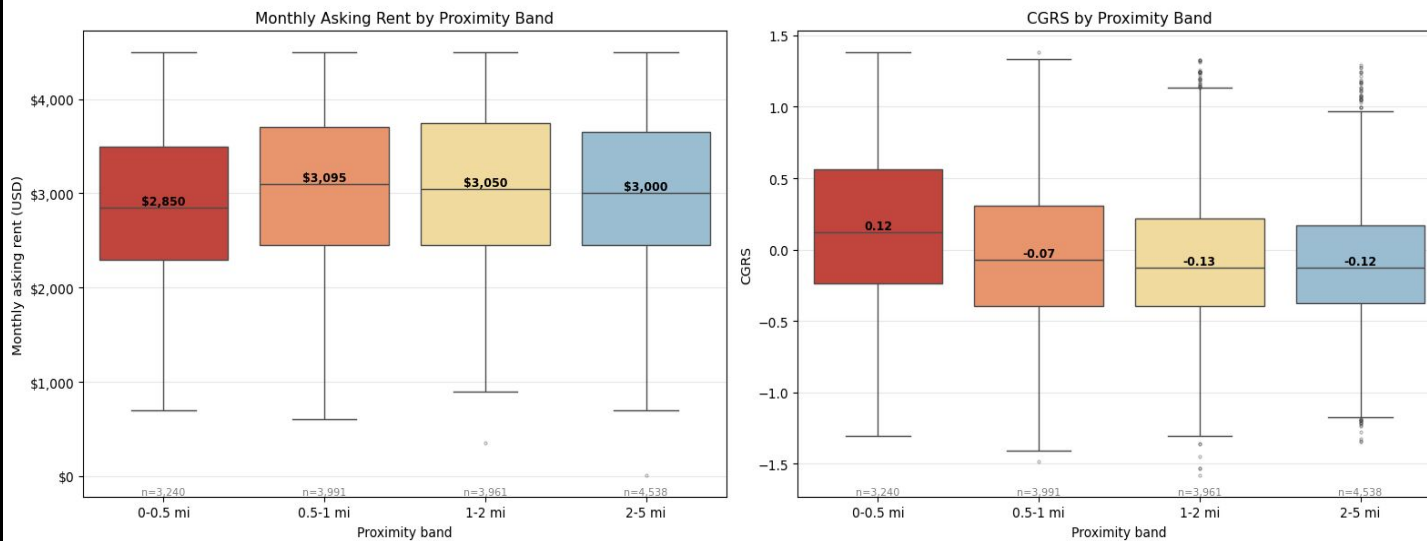
Five-fold cross-validated model performance

	CV-RMSE	CV-MAE	CV-R ²	Applicable Lines
OLS/Ridge	0.376	0.286	0.453	A E
Random Forest	0.201	0.144	0.844	A E
Gradient-Boosted Trees (best performer)	0.173	0.121	0.885	A E

Results: Graphs



Transit Proximity Effects (5-mile cap, balanced bands)

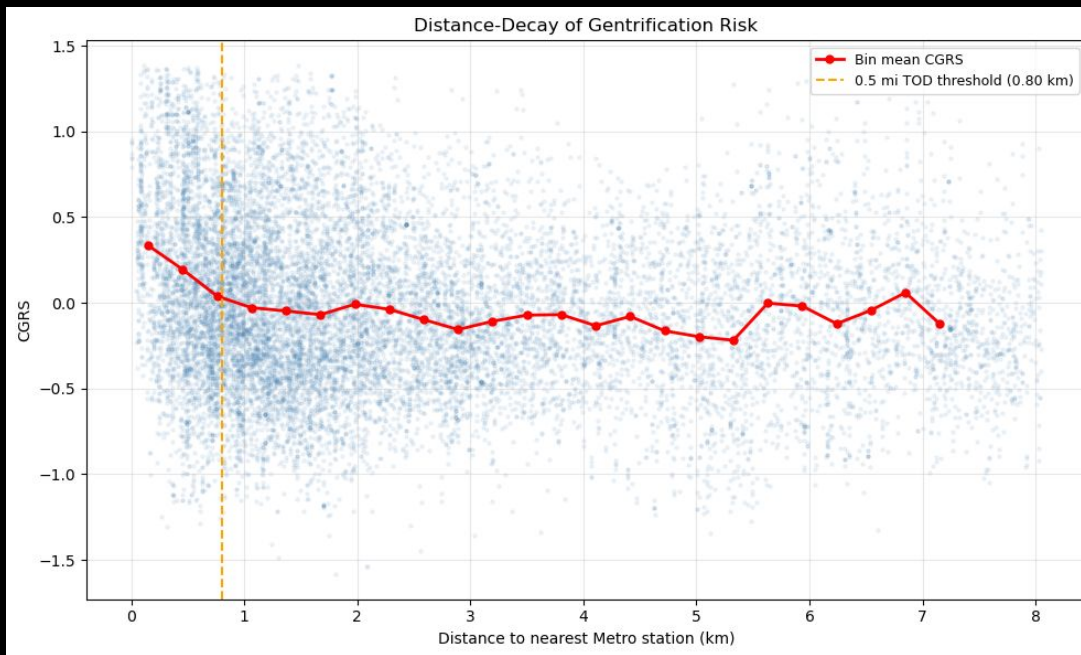
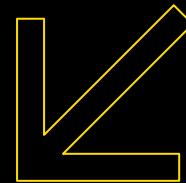


01

Transit Proximity Effects

Figure 1. Rent (left) and CGRS (right), grouped by proximity band, with sample sizes annotated for each band (5-mile cap, balanced bands).

Results: Graphs



02

Distance-Decay Plot

Distance-decay of mean CGRS, binned by distance to the nearest Metro station (25 equal-width bins from 0 to the 97th percentile of distance), with the 0.5-mile TOD threshold marked.

Implications From This Study

01

These findings are not causal. The analysis is cross-sectional, so it captures a single moment and cannot show how risk will shift as transit access expands.

02

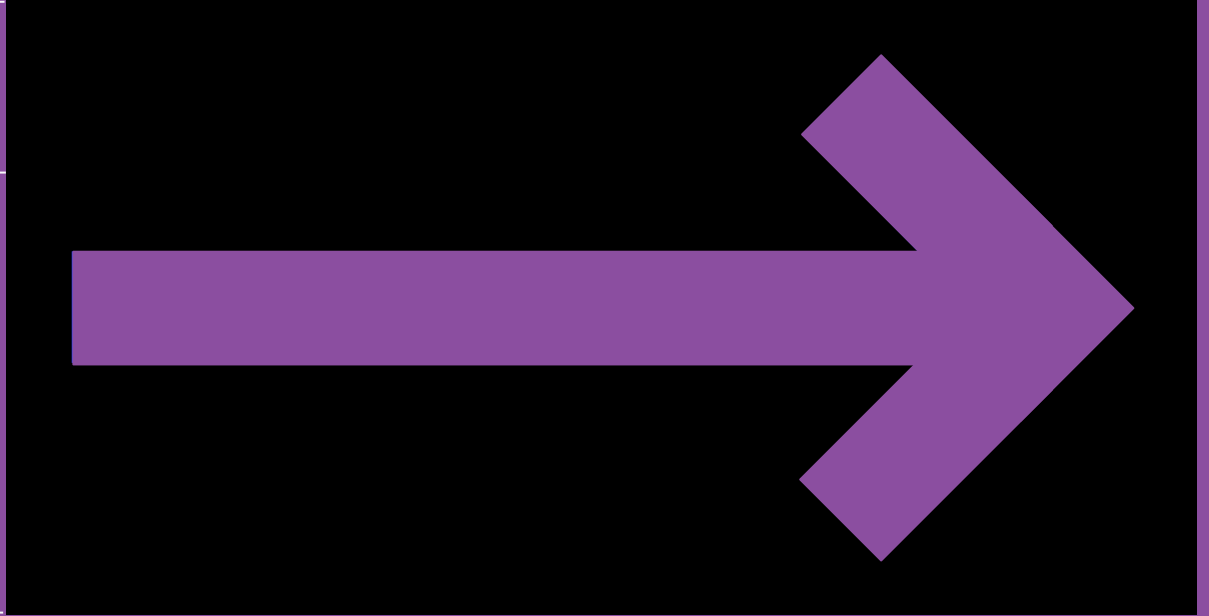
It also cannot separate the Metro's influence from the fact that LA's rail already runs through some of the city's most established corridors.

03

Proximity may reflect where high-value neighborhoods already sit, or it may be an early signal of pressure that has not yet surfaced.

Q&A

Any questions?
Ask away!



This presentation looked at the relationship between LA Metro Rail Stop Proximity and Risk of Gentrification.

Project Overview: Scope and Findings

01

Abstract

The planned D Line (Purple Line) extension to Westwood/UCLA, scheduled to open ahead of the 2028 Los Angeles Olympic Games, raises the question of whether new transit access will accelerate gentrification in the surrounding Olympic Corridor.

We compute a CGRS to address this question and isolate the predicted effect of new transit accessibility from existing neighborhood conditions.

While listings within 0.5 miles of existing Metro stations had significantly higher CGRS, this study points to a potential "ceiling effect" where additional transit access produces limited further change.



Project Objectives

01

Measure Gentrification Risk

Create a Composite Gentrification Risk Score using rent, rent appreciation, price per square foot, and education change.

02

Compare Metro Proximity

Compare rent listings near Metro stations with listing farther away to see whether closer areas show higher gentrification risk.

03

Forecast D Line Impact

Use predictive models to estimate whether the future D line Extension could add gentrification pressure to the Olympic Corridor.